6.7 Writing Two-Variable Linear Inequalities

Lesson Introduction

Benchmark	MA.912.AR.2.7 Write two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.	Learning Target	• I can write a two-variable inequality given the graph of a solution set.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.912.AR.3.9 Given a mathematical or real-world context, write two-variable, quadratic inequalities to represent relationships between quantities from a graph or a written description.

Essential Question	• How do you write an inequality given the graph of a solution set?
Lesson Narrative	In this lesson, students continue their work with two-variable inequalities and their graphs. Students will learn the process for reviewing and analyzing the graph of a linear inequality to write the inequality the graph represents. Instruction includes writing linear inequalities in different forms (slope-intercept, standard, and point-slope).

Component Summary	6.7.1 Warm-Up (~5 min)	Which One Doesn't Belong? *(SE p. 330)*	6.7.4 Guided Practice (10 min)	Practice writing linear inequalities from a graph *(SE p. 334)*
	6.7.2 Exploration (10 min)	Explore relating linear inequalities in different forms to their graphs *(SE p. 330)*	6.7.5 Activity (*see "Things to Consider" for time*)	Inequality scavenger hunt *(SE p. 335)*
	6.7.3 Guided Instruction (10 min)	Introduction to writing two-variable linear inequalities in different forms from graphs *(SE p. 332)*	6.7.6 Wrap-Up (~5 min)	Emoji reflection activity *(SE p. 336)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• Scavenger Hunt Posters from the digital teacher area.	• Place Posters around the room for 6.7.5 Scavenger Hunt.

<table>
<tr><td rowspan="4">Teacher Tips</td><td colspan="2">Common Misconceptions</td><td colspan="2">Things to Consider</td></tr>
<tr><td colspan="2">Students may struggle determining which inequality symbol to use based on the shading on the graph.
Students may be confused about which inequality symbol to use based on the boundary line types.</td><td colspan="2">6.7.5 Activity is a scavenger hunt activity. If the lesson goes too long, the scavenger hunt could be modified or moved to another place in the unit as a review activity.
Work through the activities to determine pacing that makes the most sense based on your students’ level of understanding at this point in the unit. For example, 6.7.4 Guided Practice could be assigned for homework if you think students are ready to work on this without guidance, which would allow more time to incorporate the scavenger hunt activity, if desired.</td></tr>
<tr><td colspan="2">Literacy and Language Routines</td><td colspan="2">Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td colspan="2">Group Presentations
In 6.7.5 Activity, students will work with small groups to complete a scavenger hunt activity. They will have the opportunity to share out where they started and the path they took to get to the end of the activity.</td><td colspan="2">MA.K12.MTR.3.1: Mathematical Fluency
Throughout this lesson, students are applying what they’ve learned about two-variable inequalities and their graphs. Students will select efficient and appropriate methods for solving problems within the given context in order to complete tasks accurately and with confidence.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="4">Consider using 6.7.5 Activity to work with students who may be struggling. This can be a way to provide students with additional support based on their performance.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="5">Lesson Notes:</td></tr>
</table>

6.7.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students will look at four images to make an argument for which one does not belong and explain their choice.*

6.7.1 Warm-Up: Which One Doesn't Belong?

1. Four figures are shown.

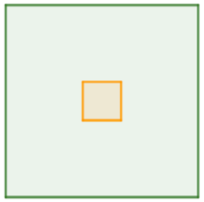


Figure 1

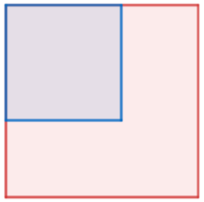


Figure 2

Figure 3

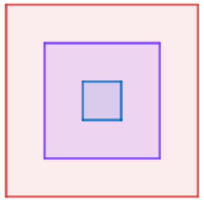


Figure 4

a. Which one doesn't belong?
Answers will vary. Figure 3.

b. Explain your choice.
Answers will vary.
Figure 1 doesn't belong because it is the only one that contains two centered squares.
Figure 2 doesn't belong because it is the only that that contains two squares that share a vertex.
Figure 3 doesn't belong because it contains a triangle, and all the other figures only have squares.
Figure 4 doesn't belong because it is the only one that includes three figures (squares).

The purpose of this activity is not to force students into a single correct answer - it is about being able to justify their choice with reasoning. There are multiple correct answers that exist for this question. Consider polling the class to see how many students selected each figure and whether their reasoning was similar or different.

6.7.2 Exploration: Relating Linear Equations to Linear Inequalities

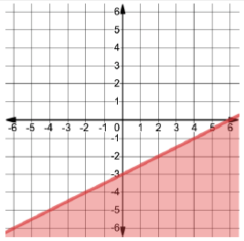
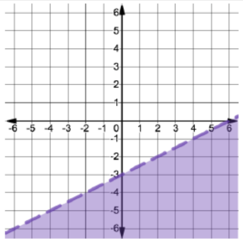
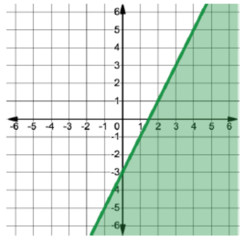
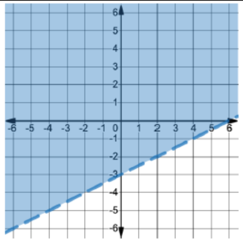
Component Overview: *Students will complete an Exploration activity where they relate linear equations to linear inequalities through two practice questions.*

6.7.2 Exploration: Relating Linear Equations to Linear Inequalities

1. Four inequalities are given.

   $y < \frac{1}{2}x - 3 \qquad y + 4 > \frac{1}{2}(x+2) \qquad y \leq 2x - 3 \qquad -x + 2y \leq -6$

   a. Work with your partner to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.

 Inequality: $-x + 2y \leq -6$	 Inequality: $y < \frac{1}{2}x - 3$
 Inequality: $y \leq 2x - 3$	 Inequality: $y + 4 > \frac{1}{2}(x+2)$

   b. Explain which graph(s) were the easiest for you to match the graph to the inequality.
   Answers will vary. The easiest ones for me to match were the inequalities in slope-intercept form.

   c. Which of the inequalities have the same boundary line?
   $-x + 2y \leq -6$
   $y < \frac{1}{2}x - 3$
   $y + 4 > \frac{1}{2}(x+2)$

Monitor students while they collaborate on this exploration with their partners to determine potential common errors they might make.

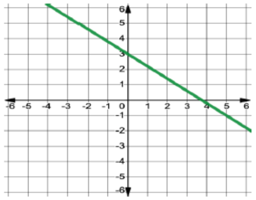
Mistakes to look out for:

- *Using the incorrect inequality sign to match the boundary line type or shaded region of the solution*
- *Writing the equation of the boundary line incorrectly*

6.7.2 Exploration: Relating Linear Equations to Linear Inequalities (continued)

d. Explain how you distinguished between the graphs of the inequalities with the same boundary line using the different forms of the inequalities given.
Answers will vary. To distinguish between the graphs of the inequalities with the same boundary lines, I converted all the inequalities to slope-intercept form and then used the inequality symbols to determine which matched each graph.

2. The graph of a two-variable linear equation is shown.

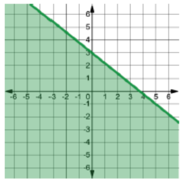


a. Write the equation of the line in slope-intercept form.

$y = -\frac{4}{5}x + 3$

b. Suppose the line shown on the coordinate grid is the boundary line of a two-variable linear inequality. Discuss with your partner which parts of the inequality would be the same as the equation from Part A, and which parts of the inequality would be different.
Answers will vary. The only thing that would be different is that the equal sign would be an inequality symbol.

c. The graph shows the solution set of an inequality with the same line as the boundary line.



Write the inequality represented on the graph in slope-intercept form.

$y \leq -\frac{4}{5}x + 3$

How did you determine which inequalities have the same boundary line?

At the conclusion of the Exploration, guide students through a discussion of their answers to question 2b. This can make for a rich mathematical discussion, as well as address any potential misconceptions.

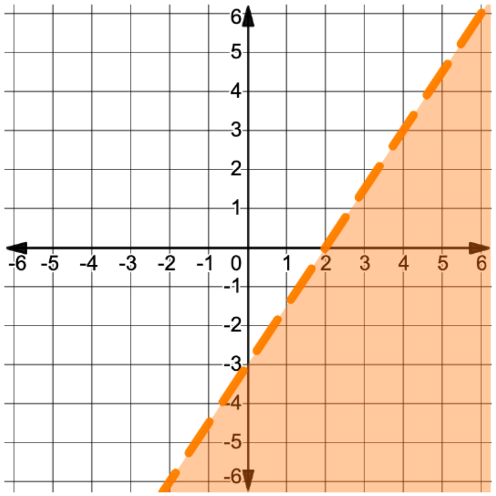
6.7.3 Guided Instruction: Writing Two-Variable Linear Inequalities in Different Forms

Component Overview: *Students will be introduced to writing a two-variable, linear inequality in different forms given a graph.*

6.7.3 Guided Instruction: Writing Two-Variable Linear Inequalities in Different Forms

Similar to the equations of lines, two-variable linear inequalities can be written in slope-intercept, point-slope, or standard form.

1. The graph of a two-variable linear inequality is shown.



a. Write the inequality of the solution set graphed above in **slope-intercept form**.

$y < \frac{3}{2}x - 3$

b. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

Answers will vary. The inequality symbol is $>$ or $\geq$ if the graph is shaded above the boundary line and $<$ or $\leq$ if the graph is shaded below the boundary line. The symbol is $\geq$ or $\leq$ if the boundary line is solid.

Draw attention to the bolded words in question 1a. There will be opportunities throughout the lesson to write the inequalities in different forms.

2. Consider how to use the graph to write the inequality in **point-slope form**.

   a. Write an equation for the boundary line in point-slope form.
   Answers will vary.
   $y - 3 = \frac{3}{2}(x - 4)$

   b. Select a test point on one side of the boundary line. Substitute the x- and y-value of the test point into the equation of the boundary line in point-slope form and evaluate. Show your work.
   Answers will vary.
   $1 - 3 = \frac{3}{2}(4 - 4)$

   c. Use your work from Part B to determine which inequality symbol to use. Then, write an inequality in point-slope form for the graphed solution set.
   Answers will vary.
   $y - 3 < \frac{3}{2}(x - 4)$

   d. Explain why there are many different point-slope form inequalities that could have the same solution set.
   Answers will vary. You could choose any point along the boundary line to write the point-slope form inequality, so there are infinitely many inequalities that would match the solution set.

If students do not remember how to write the equation in point-slope form, refer to Unit 3 Lesson 2.

Why does the point used need to be on the boundary line to write the equation in point-slope form?

6.7.3 Guided Instruction: Writing Two-Variable Linear Inequalities in Different Forms (continued)

3. Eli and Mariska each wrote an inequality in **standard form** for the same graphed solution set. Their inequalities and reasoning are shown.

Eli	Mariska
"I took the slope-intercept form inequality and re-wrote it in standard form. First, I multiplied the inequality by 2 to eliminate the fractional coefficient, and then I added $-3x$ to both sides of the inequality as shown." $y < \frac{3}{2}x - 3$ $2y < 3x - 6$ $-3x + 2y < -6$	"I looked at the intercepts of the boundary line, $(0, -3)$ and $(2, 0)$. I know that 18 is a common multiple of -3 and 2, so I wrote an equation of the boundary line set equal to 18. Then, I used the intercepts to determine the coefficients of each term in the equation $9x - 6y = 18$. Lastly, I used a test point to determine the inequality symbol to use, and I got $9x - 6y > 18$."

a. Discuss with your partner whether each student is correct or incorrect. Summarize your discussion.
Answers will vary. Both students are correct. The inequalities are equivalent since if you multiply both sides of Eli's inequality by -3, you would get Mariska's inequality.

b. Write another inequality in standard form that would have the same solution set.
Answers will vary.
$-15x + 10y < -30$

Given the graph of a solution set, there are **infinitely many** inequalities that could be written with that solution set. The inequality can be written in slope-intercept form, point-slope form, or standard form.

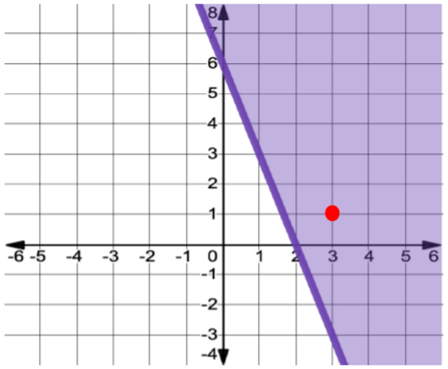
Consider having students use graphing technologies and prior learning from Units 3, 4, and 5 to show that the two inequalities are equivalent, since both students' strategies work.

6.7.4 Guided Practice: Writing Inequalities Given the Graph of the Solution Set

Component Overview: *Students will complete two practice questions with writing inequalities given the graph of the solution set.*

6.7.4 Guided Practice: Writing Inequalities Given the Graph of the Solution Set

1. The solution set of an inequality is shown on the coordinate grid.



Write an inequality in each form given the solution set shown on the graph. Check your work by substituting a test point into each inequality.

Test points used will vary. Example uses $(3,1)$.

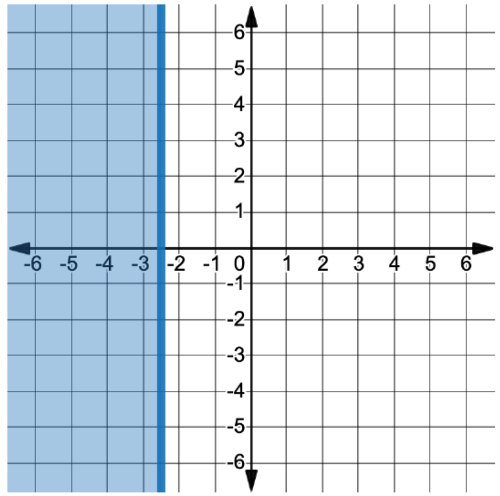
	Inequality	Check With Test Point
Slope-Intercept Form	$y \geq -3x + 6$	*Answers will vary.* $1 \geq -3(3) + 6$ $1 \geq -3$
Point-Slope Form	*Answers will vary.* $y - 3 \geq -3(x - 1)$	*Answers will vary.* $1 - 3 \geq -3(3 - 1)$ $-2 \geq -6$
Standard Form	*Answers will vary.* $3x + y \geq 6$	*Answers will vary.* $3(3) + 1 \geq 6$ $10 \geq 6$

Consider leveraging a Think-Pair-Share instructional routine with question 1. Allow students a few minutes to try to complete the table on their own, and then compare their answer with a partner to determine if their inequalities are accurate and their test points work.

Consider assigning this component for homework if your observations indicate students would be successful without much guidance. This would allow more time to use the scavenger hunt activity as practice in this lesson.

6.7.4 Guided Practice: Writing Inequalities Given the Graph of the Solution Set (continued)

2. The solution set of an inequality is graphed below.



a. Which of the following inequalities represents the solution set graphed above?

- ○ $y \geq -2.5$
- ○ $y \leq -2.5$
- ○ $x \geq -2.5$
- ○ $x \leq -2.5$

b. Explain your selection.
Answers will vary. I chose $x \leq 2.5$ because it is a vertical line and if I look at the x-axis, I see that the numbers that are in the shaded region are less than 2.5.

For students who get stuck on question 2, refer them back to their notes from the prior lesson.

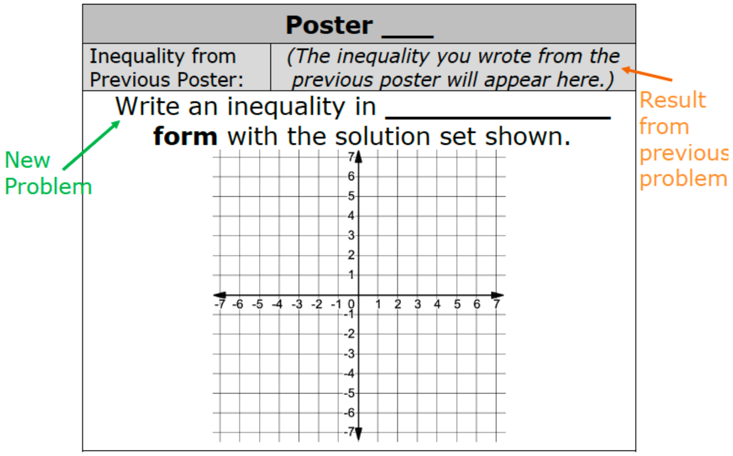
How can you verify your answer is correct using a test point?

6.7.5 Activity: Scavenger Hunt

Component Overview: *Students will complete an inequality scavenger hunt using the provided poster templates, each with the graph of an inequality.*

6.7.5 Activity: Scavenger Hunt

Work with your partner or group to complete the scavenger hunt. Each poster shows the following information.



- Write the letter of the poster you start with in the first row of the table. Then, work with your group or partner to write an inequality for the graphed solution set shown on the poster in the form indicated. Verify that your inequality correctly matches the solution set using a test point.
- Look for the card that matches your inequality in the grey area to find your next problem.
- Repeat this process until your group has completed all the scavenger hunt problems and end up back at your original problem.

For this activity, there is some preparation work that will need to be completed prior to the launch of the activity. Post the posters from the activity document around the room.

At the end of this activity, students will know if their work is correct if they went to the posters in the correct order.

6.7.5 Activity: Scavenger Hunt (continued)

Poster	Inequality in the Form Indicated	Verify With a Test Point
K	$y > 3$	Answers will vary. $4 > 3$
R	$3x + 4y \leq 20$	Answers will vary. $3(1) + 4(1) \leq 20$ $7 \leq 20$
B	$y \leq 3x - 4$	Answers will vary. $1 \leq 3(2) - 4$ $1 \leq 2$
L	$x > 3$	Answers will vary. $4 > 3$
F	$y - 2 > -\frac{1}{3}(x - 1)$	Answers will vary. $3 - 2 > -\frac{1}{3}(3 - 1)$ $1 > -\frac{2}{3}$
W	$y \leq -\frac{3}{4}x - 1$	Answers will vary. $-3 \leq -\frac{3}{4}(1) - 1$ $-3 \leq -1\frac{3}{4}$

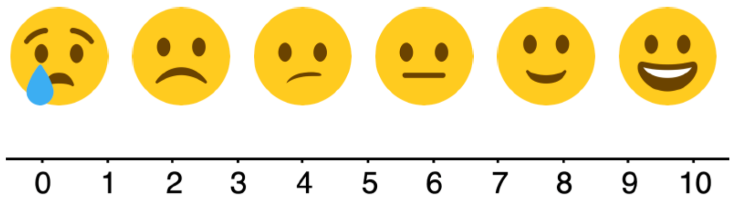
The answer order can vary since students are able to start with any poster, but the letters that follow should be the same.

6.7.6 Wrap-Up: Reflection

Component Overview: *Students will use emojis to describe how confident they feel in writing a two-variable inequality in different forms when given a graph.*

6.7.6 Wrap-Up: Reflection

1. Use the emoji scale to describe how confident you feel in writing a two-variable inequality in different forms when given the graph of a solution set.
   Answers will vary.



2. Explain your selection.
   Answers will vary.

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.